Akhil Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
RJ College of Arts, Science and Commerce	Class XII (Maharashtra board of higher secondary education)	2010	79.17%
AVM High School	Class X (Maharashtra board of higher secondary education)	2008	82.46%

EXPERIENCE

IIT Hyderabad

Research Assistant

Worked on SAR ADC for the Transceivers IC.

Shelby Foods LLP

Co-Founder and Chief Operating Officer(COO)

- Led **day-to-day business operations**, overseeing production, procurement, inventory, logistics, and supply-chain activities.
- Developed and implemented **operational strategies and processes** to improve efficiency, productivity, and cost management.

PROJECTS

Master Thesis: Design of Transceiver for On-Board Wireless Interface of Distributed Control System using TSMC 65-nm Technology

Thesis Advisor: Dr. Abhishek Kumar

- Designed an **8-bit, 20-MSPS SAR ADC** in TSMC 65-nm CMOS with a 1.2-V supply and 100-µW power consumption.
- Achieved 7.93-bit ENOB through transient-noise analysis, with DNL of 0.30 to +0.10 LSB and INL of 0.263 to +0.183 LSB.
- Characterized dynamic performance at an input frequency of Fs * 7/64, achieving 48-dB SINAD and 64-dB SFDR; calculated a Walden FoM of ~19.5 fJ/conversion-step.
- Verified ADC operation across PVT corners with limited ENOB variation and performed Monte Carlo mismatch analysis on the comparator.

Analog Circuit Design

Two-Stage Miller-Compensated Op-Amp (TSMC 180 nm)

- Designed and simulated a two-stage CMOS op-amp with Miller compensation for high gain and stability

Fully Differential Folded Cascode Amplifier based buffer to drive the SAR ADC

- Developed a folded-cascode amplifier achieving **~52 dB** gain to drive the SAR ADC.
- Implemented **CMFB** for stable differential operation and verified performance in Cadence Spectre.

CMOS Image Sensor Design -- Course Project

- Designed and simulated **3T** and **4T CMOS** active-pixel sensor (APS) circuits and implemented a pixel array for image-sensing applications.
- **Designed a Correlated Double Sampling (CDS)** readout block to reduce pixel offset and fixed-pattern noise.
- Implemented supporting mixed-signal blocks including an 8-bit single-slope ADC, ramp generator, comparator, and counter for pixel readout and digitization.
- Verified the functionality of the pixel array and readout chain through transient-level simulations in Cadence Virtuoso.

RFIC Project:

Negative-Resistance Amplifier for Intelligent Reflecting Surface (IRS) Antenna Array

- Designed and simulated a **5-GHz negative-resistance amplifier** for an intelligent reflecting surface (IRS) antenna array with a **50- RF interface**.
- Achieved **9.8-dB reflection gain (S11)** at the target frequency and a **1-dB compression point (P1dB) of approximately 8 dBm**.
- Analyzed the circuit's **negative input resistance, reflection characteristics, and large-signal performance** for reflection-amplification applications.

SCHOLASTIC ACHIEVEMENTS

- Secured **A Grade** in Nanoelectronics, CMOS Image Senosrs and, Semiconductor Devices and Modelling at IIT Hyderabad.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) for **NPTEL** course: **IC design for Wireless Communication**

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

Teaching Assistant (TA): **IC Design for Wireless Communication** Course for 4th-year UG and 1st-year PG students

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

Teaching Assistant (TA): **Analog IC Design** Course for 3rd-year UG and 1st-year PG students

Managed key course logistics. **Instructor:** Dr. Abhishek Kumar

MMA Club IIT Hyderabad

General Secretory and Part time Wrestling Instructor

SKILLS

Tools: Cadence (Virtuoso, Incisive/Xcilium, Genus, Innovus), Xilinx Vivado, ADS, KiCad, MATLAB

RELEVANT COURSES

- **Analog Domain** : RFIC, Analog IC Design, Analog IC Lab, CMOS Image Sensor.
- **Digital Domain** : Introduction to VLSI Design, Digital IC Design, Digital IC Design Lab, System Design Lab.
- **Memory Domain** : Memory Circuits and Systems .
- **Device Domain** : Semiconductor Device Modelling, VLSI technology.

EXTRACURRICULAR

Boxing, Wrestling, Chess and Horse Riding